

ACETAMOL 650MG TABLET

Composition:

Each film coated tablet contains 650mg Paracetamol

Description:

A white to off white oblong shaped film coated tablet.

Pharmacodynamics:

Paracetamol is an antipyretic analgesic. Paracetamol is an effective analgesic and antipyretic but lacks anti-inflammatory action. The mechanism of action is probably similar to that of aspirin and dependent on the inhibition of prostaglandin synthesis. This inhibition appears, however to be on a selective basis.

Analgesic-The mechanism of analgesic action has not been fully determined. Paracetamol may act predominantly by inhibiting prostaglandin synthesis in the central nervous system (CNS) and, to a lesser extent, through a peripheral action by blocking pain-impulse generation. The peripheral action may also be due to inhibition of prostaglandin synthesis or to inhibition of the synthesis or actions of other substances that sensitize pain receptors to mechanical or chemical stimulation.

Antipyretic- Paracetamol act centrally on the hypothalamic heat-regulating center to produce peripheral vasodilatation resulting in increased blood flow through the skin, sweating and heat loss. The central action probably involves inhibition of prostaglandin synthesis in the hypothalamus.

Pharmacokinetics:

Absorption

Paracetamol is rapidly and almost completely absorbed from the gastrointestinal tract. The concentration in plasma reaches a peak in 30 to 60 minutes and the plasma half-life is 1 - 4 hours after therapeutic doses. Oral bioavailability is 60 to 98%. Absorption half-life 4.5 minutes with no lag time.

Distribution

Paracetamol is relatively uniformly distributed throughout most body fluids. Binding of the drug to plasma proteins is variable; 20 to 30% may be bound at the concentrations encountered during acute intoxication.

Metabolism

Paracetamol is metabolized by the hepatic microsomal enzyme system. In the overdose situation when the sulfate and glucuronide stores are saturated, a large percentage of the dose is oxidized to cysteine and mercapturic acid conjugates. In adults, majority of paracetamol is conjugated with glucuronide (inactive) and to a lesser extent, with sulfate (inactive). In premature infants,

newborns and young infants, majority of paracetamol is conjugated with sulfate. In adults at therapeutic doses, paracetamol is mainly conjugated with glucoronide (45 to 55%) or sulfate (20 to 30%). A minor proportion (less than 20%) is metabolized to catechol derivatives and mercapturic acid compounds via oxidation. A minor hydroxylated metabolite (N-acetyl-p benzoquinoneimine), is usually produced in very small amounts by cytochrome P450 isoenzymes (mainly CYP2E1 and CYP3A4) in the liver and kidney. It is usually detoxified by conjugation and glutathione but may accumulate after paracetamol overdosage and cause tissue damage.

Excretion

Paracetamol excretion is rapid and occurs mainly via the urine. Paracetamol's metabolic products are excreted by the kidney in urine mainly as the glucoronide and sulfate conjugates. Following therapeutic doses 90 - 100% of the drug may be recovered in the urine within the first day. However, practically no paracetamol is excreted unchanged and the bulk is excreted after hepatic conjugation. 1% to 4% paracetamol excreted unchanged (as free paracetamol). The half-life may be prolonged in acute overdose or severe liver disease.

Indication:

ACETAMOL 650mg is taken for the relief of mild to moderate pain of painful disorders such as headache, backache, period pain and aches due to colds and flu, pain related to mild arthritis and the relief of fever.

Recommended Dosage:

Adults: 1-1 1/2 tablets every 4 to 6 hours as required (not more than 6 tablets in a day unless prescribed by doctor). Do not exceed the recommended dose. Do not continue taking this drug for more than 10 days for pain or 3 days for fever.

Route of Administration:

Oral administration

Contraindication:

Hypersensitivity to paracetamol.

Warning and precautions:

This preparation contains paracetamol.
Do not take any other paracetamol containing medicines at the same time.

Paracetamol provides symptomatic relief only, additional therapy to treat the cause of the pain or fever should be instituted when necessary. Patients should contact their health care provider if the pain lasts for more than 10 days or fever lasts more than 3 days. Paracetamol should be given with care to patients with impaired kidney or liver function. Large doses should be avoided in

patients with hepatic impairment. It should be given with care to patients with alcohol dependence.

Paracetamol overdose may harm the liver. Follow the recommended dose.

Allergy alert: Paracetamol may cause severe skin reactions. Symptoms may include skin reddening, blisters or rash.

These could be signs of serious condition. If these reactions occur, stop use and seek medical assistance right away.

Interaction with other medicaments:

The risk of paracetamol toxicity may be increased in patients receiving other potentially hepatotoxic drugs e.g. antibacterials like isoniazid, antivirals like zidovudine (may also cause neutropenia) and interferon alfa or drugs that induce liver microsomal enzymes (which may also reduce paracetamol effectiveness) e.g. antibacterials like rifampicin, antiepileptic drugs such carbamazepine, Phenobarbital, phenytoin or primidone.

The absorption of paracetamol may be accelerated/enhanced by drugs which increase gastric emptying such metoclopramide and domperidone.

Excretion may be reduced and plasma concentrations altered when given with probenecid.

Colestyramine (lipid regulating drug) may reduce the absorption of paracetamol if given within 1 hour of paracetamol.

The rate of paracetamol absorption may be decreased by drugs which decrease gastric emptying (e.g. propantheline, or narcotic analgesics, pethidine)

Paracetamol theoretically inhibits metabolism of intravenous busulfan (a cytotoxic drug).

Paracetamol should be minimized or eliminated less than 72 hours before the administration of busulfan.

Paracetamol theoretically decreases lamotrigine effectiveness by increasing renal clearance.

Alcohol

Concurrent use of alcohol and paracetamol may increase the metabolism of paracetamol to the highly hepatotoxic metabolite NAPQI and increase risk of hepatotoxicity. Caution should be used with patients who drink 3 or more alcoholic beverages per day and take paracetamol. Patients should be advised not to exceed 4 grams of paracetamol per 24 hours. Chronic alcoholics should avoid the use of paracetamol. In cases of paracetamol overdose consider treatment with acetylcysteine.

Warfarin

The concurrent use of paracetamol has been associated with an increased hypoprothrombinemic effect of warfarin leading to increased risk of bleeding. The probable mechanism is inhibition of warfarin metabolism or interference with clotting factor formation. Patients receiving warfarin or other coumarin anticoagulants should be cautioned to limit their intake of paracetamol. Elevations in INR have occurred within 1-2 weeks of initiating paracetamol at moderate to high

doses (2 and 4g/day) in patients on stable warfarin therapy. Consider early (4 days after starting paracetamol) and frequent monitoring of INR for several weeks when paracetamol is added or discontinued in patients on warfarin therapy.

Pregnancy and lactation:

Pregnancy

FDA Category B: Either animal-reproduction studies have not demonstrated a fetal risk but there are no controlled studies in pregnant women, or animal-reproduction studies have shown an adverse effect (other than a decrease in fertility) that was not confirmed in controlled studies in women in first trimester (and there is no evidence of a risk in later trimesters).

Australian Drug Evaluation Committee's (ADEC) Category A: Drugs which have been taken by a large number of pregnant women and women of childbearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the fetus having been observed.

Although it crosses placenta, paracetamol is considered to be safe in normal therapeutic doses for short-term use as a minor analgesic/antipyretic in pregnancy. It is generally regarded as the drug of choice for use in these settings. Overdose or prolonged use of high therapeutic doses may result in fetal liver toxicity and other adverse effects.

Lactation

World Health Organization Rating: Compatible with breastfeeding The weight of an adequate body of evidence and/or expert consensus suggests this drug poses minimal risk to the infant when used during breastfeeding.

Paracetamol is considered compatible with breastfeeding by the American Academy of Pediatrics.

Although paracetamol appears in breast milk, the concentration does not appear sufficient to result in pharmacologically significant dose to the breastfeeding infant. The drug is well tolerated when given directly to the infants

Side effects:

When taken as directed at therapeutic doses, paracetamol is virtually free of side effects. However, rare side effects such as urticarial or erythematous skin rash, hypersensitivity reactions, drug fever and mucosal lesions, thrombocytopenia, leucopenia, pancytopenia, neutropenia and agranulocytosis have been reported. Overdosage with paracetamol can result in severe liver damage and sometimes acute renal tubular necrosis.

Cutaneous hypersensitivity reactions including skin rashes, angioedema, Stevens Johnson Syndrome/ Toxic Epidermal Necrolysis have been reported.

Overdosage:

- Adults taking as little as 10 to 15 g of paracetamol within 24 hours may develop severe hepatocellular necrosis and, less often, renal tubular necrosis.
- Patients should be considered at risk of severe liver damage if they have ingested more than 150 mg/kg of paracetamol or 12 g or more in total, whichever is the smaller.

Symptoms

- Early signs of overdosage (very commonly nausea and vomiting although they may also include lethargy and sweating) usually settle within 24 hours.
- Abdominal pain may be the first indication of liver damage, which is not usually apparent for 24 to 48 hours and sometimes may be delayed for up to 4 to 6 days after ingestion.
- Liver damage is generally at a maximum 72 to 96 hours after ingestion. Hepatic failure, encephalopathy, coma, and death may result.

Treatment:

Activated charcoal may be used to reduce gastrointestinal absorption, if it can be given within 1 hour of the overdose, and if more than 150 mg/kg or 12 g of paracetamol (whichever is smaller) has been ingested. However, if antidote is to be given orally the charcoal is best cleared from the stomach to prevent it reducing the absorption of the antidote. The stomach should be emptied by lavage or emesis with syrup of ipecac. Antidote treatment should be started as soon as possible after suspected overdosed paracetamol ingestion and should not be delayed while awaiting the result of plasma assays. Acetylcysteine is usually the antidote of choice.

Acetylcysteine dosage:**IV****Adult :**

An initial dose of 150 mg/kg (maximum of 16.5 g) of acetylcysteine in 200 mL of glucose 5% is given intravenously over 15 minutes in the UK, or over 60 minutes in the USA. This is followed by an intravenous infusion of 50 mg/kg (maximum of 5.5 g) in 500 mL of glucose 5% over the next 4 hours. Then, 100 mg/kg (maximum of 11 g) in one litre over the next 16 hours. Sodium chloride 0.9% may be used where glucose 5% is unsuitable. The volume of intravenous fluids should be modified for children. If an anaphylactoid reaction develops, the infusion should be stopped and an antihistamine given; it may be possible to continue the acetylcysteine infusion at a slower rate. Children:

AMH: Treat children with the same dosage regimens, but give the initial 150mg/kg dose in 5mL/kg glucose 5%, the 50mg/kg maintenance dose in 10ml/kg and the final 100mg/kg in 20mL/kg in order to avoid fluid overload and hyponatremia.

Oral

Adult: - In the USA, acetylcysteine may also be given orally as an alternative to parenteral treatment. It is given as an initial dose of 140 mg/kg as a 5% solution followed by 70 mg/kg every 4 hours for an additional 17 doses. - Any dose vomited within 1 hour of administration must be repeated. If necessary, the antidote may be given via duodenal intubation.

Supportive treatment consists of maintaining fluid and electrolyte balance, correcting hypoglycemia and administering vitamin K1 (if prothrombin time ratio exceeds 1.5) and fresh frozen plasma or clotting factor concentrate (if prothrombin ratio exceeds 3.0)

Effects on Ability to Drive and Use Machine:

None known.

Presentation:

10 tablets in a blister. 3x10's, 8x10's, 10x10's, 50x10's in a box.

Storage Conditions:

Store below 30°C.

Product Registration Holder and Manufacturer:

Noripharma Sdn. Bhd.
Lot 5030, Jalan Teratai,
5 1/2 Mile off Jalan Meru,
41050 Klang, Selangor
Malaysia

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